J. Cosmet. Sci., 53, 375–386 (November/December 2002)

Stability and release of topical tranexamic acid liposome formulations

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Accepted for publication April 29, 2002.

Synopsis

Tranexamic acid (TA) has been claimed to have whitening effects. The effects of TA contents (5% and 10%) and charges on the stability and release of TA entrapped in hydrogenated soya phosphatidylcholine/cholesterol/charged lipid {dicetyl phosphate (–) or stearylamine (+)} liposomes at molar ratios of 7:2:1(–) and 7:2:1 (+) were investigated. The TA contents were determined spectrophotometrically at 415 nm, following derivatization with 2,4,6-trinitrobenzosulfonic acid. Stability and leakage of TA from liposomes were characterized at 4°, 30° and 45°C for 90 days. The leakage rates of TA in negative liposomes were lower than those in positive liposomes. The TA in all liposome formulations was relatively stable, as > 90% of total drug remained after up to two months. The release of TA from liposomes was examined using vertical Franz diffusion cells at 37°C for 24 h. The release rates of TA from all liposome formulations were ~ 3 times lower than those from solutions. Charges appeared to affect the physical stability, leakage, and shelf life of TA in liposomes, whereas TA concentrations seemed to affect the release of TA. The 7:2:1 (10% TA,–) liposome was the best formulation, due to its small size, low leakage, high stability, and prolonged and sustained release profile.

INTRODUCTION

Liposomes are excellent novel formulations as drug and cosmetic carriers, owing to their biodegradability, biocompatibility, low toxicity, and ability to entrap lipophilic and hydrophilic drugs (1,2). In addition, liposomes may enhance the penetration of drugs into the skin with a slow release and have a moisturizing effect (3). The tissue distribution and release of drugs from liposomes may be affected by particle size, composition of the lipid, and surface charges of the liposomes (4). In scintigraphic studies, liposomes carrying radionuclides within the aqueous space have potential diagnostic imaging applications (2). Gamma-emitting radionuclides, such as ^{99m}Tc, ^{111}In and ^{67}Ga have

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been encapsulated in liposomes for imaging agents (5–7). Some clinical trials for tumor imaging using labeled liposomes have already been initiated (7–9).

Tranexamic acid (TA) is a hydrophilic drug used as an antifibrinolytic agent (10–15). It is a *trans*-4-aminomethyl cyclohexane carboxylic acid ($C_8H_{15}NO_2$, mw 157.21) (16). It has been claimed that TA has anti-inflammatory (16) and whitening effects for topical use (17). The current commercially available preparations of TA are tablets and injections (18). We have previously developed liposome formulations for TA from various lipid compositions, containing neutral or positively or negatively charged lipids, which may potentially reduce the irritation and allergy caused by TA and improve the moisturizing effect (19). It was found that the charged liposome composed of hydrogenated soya phosphatidylcholine/cholesterol/stearylamine (+) or dicetyl phosphate (−) at a molar ratio of 7:2:1 for TA demonstrated 13–16% of drug entrapment. Thus, these liposome formulations were selected for further stability and release studies. It is expected that the developed liposome formulations can be applied for the future development of TA, not only as a sustained release preparation but for topical whitening cosmetics as well.

The present study reports the characterization of stability and release of TA from various multilamellar liposome formulations, prepared by a chloroform film method with sonication. The release rates of TA from liposomes were compared with those of free TA from solutions (5% and 10% in deionized water). Various factors (TA content and charges of liposomes) that may affect the stability and release properties of TA are discussed.

EXPERIMENTAL

MATERIALS

TA was obtained commercially from Asahi Chemical Industry Co., Ltd. (Japan). Hydrogenated soya phosphatidylcholine (Emulmetik 90®) (HSC) was kindly donated by JJ-Degussa (T) Ltd., Bangkok. Cholesterol (CHL), dicetyl phosphate (DCP), stearylamine (SA), 2,4,6-trinitrobenzosulfonic acid, and boric acid were purchased from Sigma Chemical Company (St. Louis, MO). Triton-X 100 was obtained from BDH Ltd. (Poole, England). Chloroform and potassium dihydrogen orthophosphate were of analytical reagent grade.

PREPARATION OF LIPOSOMES

Liposome dispersion samples were prepared by a chloroform film method with sonication as previously described (19). The lipids used were neutral (HSC and CHL) and charged lipids (SA for positively charged lipids and DCP for negatively charged lipids), with a total lipid concentration of 25 mg/ml. Liposome formulations composed of HSC/CHL/SA = 7:2:1(+) and HSC/CHL/DCP = 7:2:1(−) at molar ratios, with the entrapped TA (5% and 10% solutions in DI water), were prepared. For each lot, an amount of 60 ml of each liposome dispersion sample was prepared. The liposome dispersion samples were kept at 4° ± 1°C and protected from light prior to use.

LIPOSOME SIZE DETERMINATION

The particle size and size distribution of liposome dispersion samples were measured by a light-scattering particle analyzer (Mastersizer S Long Bed Ver. 2.11, Malvern Instruments Ltd., Malvern, UK), ten days after sample preparation (kept at 4°C). The particle size range was set between 0.05 and 800 μm, with the beam length at 2.40 mm and the dispersant refractive index at 1.3300. Polydisperse model analysis was employed.

STABILITY STUDY

Each liposome formulation (5 ml for physical and 20 ml for chemical stability studies) was transferred into a vial with stopper prior to storage at three different temperatures, namely 4°, 30°, and 45° (± 1°)C for 90 days. At predetermined time intervals (0, 14, 30, 60, and 90 days), 1.5 g of sample was withdrawn and mixed with 1.5 g of DI water, then centrifuged at 150,000 g (4°C) for 90 min in a Centrikron T-1180 ultracentrifuge (Kontron Instruments, Milan, Italy). The supernatant was removed and diluted (100 times) with DI water prior to the determination of the amount of the unentrapped TA. The pellet was dissolved in 5 ml of 10% Triton-X 100 solution and sonicated for 20 min. This solution was further diluted (ten times) with Triton-X 100 solution prior to the determination of the amount of entrapped TA in liposomes. This study was performed to estimate the leakage rate of entrapped TA from various liposome formulations.

To characterize the chemical stability of total TA in liposomes, samples (0.25 g) withdrawn at predetermined time intervals (see above) were dissolved in 12.25 ml of 10% Triton-X 100 solution and sonicated for 20 min. These solutions were further diluted (3.33 times) with Triton-X 100 solution prior to the determination of the total amount of TA in liposomes. An amount of 0.1 ml of the above samples (unentrapped, entrapped, and total TA) was withdrawn and derivatized, following the procedures described in the Analytical Method section below, prior to analysis. This study was conducted in six replicates.

RELEASE STUDY

Various liposome formulations, 5% and 10% TA solutions in DI water, were used as samples in the release study. The vertical Franz diffusion cells (Crown Bio Scientific, Inc., Somerville, NJ) were set at 37° ± 1°C, and the receiver chamber was filled with 12 ml of DI water. A molecular porous membrane tubing with molecular weight cutoff of 12,000–14,000 was fixed in the diffusion cell (contact area, 1.77 cm^2). The membrane was soaked prior to use overnight in DI water. Each liposome formulation (3.0 g) was centrifuged at 150,000 *g* (4°C) for 90 min in an ultracentrifuge, and the supernatant was discarded. The pellet containing liposomes was resuspended in 6.0 ml of DI water, and the dispersion sample (2.0 ml) was loaded into the donor chamber. At predetermined time intervals (0.25, 0.5, 1, 2, 4, 6, 8, and 24 h), 0.5 ml of sample in the receiver chamber was withdrawn and then assayed spectrophotometrically for TA content, following the procedures described in the Analytical Method section below. After each sampling, 0.5 ml of DI water was added to the receiver chamber to replace the loss of receiver medium. The correction factor of this dilution was used to calculate the TA concentration of the next sample. This study was performed in six replicates.

ANALYTICAL METHOD

The concentration of TA in samples was determined spectrophotometrically at 415 nm (Milton Roy Spectronic 1001 Plus, Rochester, NY), following derivatization with 2,4,6-trinitrobenzosulfonic acid (20). The color reagent employed in this study was 1.68% (w/v) 2,4,6-trinitrobenzosulfonic acid solution in DI water, and was freshly prepared and protected from light. Each 0.1 ml of the working standard solution or samples was spiked with 0.25 ml of 0.025 M disodium tetraborate solution (pH 10) and 0.25 ml of color reagent, prior to standing at 25°C for 30 min. The solution was then diluted to 5.0 ml with 0.1 M potassium dihydrogen phosphate solution (pH 4.5) to terminate the reaction. The formed color intensity was relatively stable in the reaction medium for at least 3 h when protected from light. The solution mixture in the absence of TA was used as a blank. A calibration graph was constructed.

DATA ANALYSIS

The proposed models (zero order, first order, and Higuchi models) were tested by fitting the experimental data to the appropriate equations (Table I). The correlation was used as an indicator of goodness-of-fit of the equation to the experimental data. The leakage, degradation, and release rate constants (k) of TA from the liposomes were estimated from the slope of TA concentration versus time plots, by least-squares fitting of the rate equation (Table I). The predicted shelf life (t_{90}), namely the time required when the entrapped TA content in the liposome remains at 90%, was estimated by substituting k into the shelf life equation for first-order kinetics (Table I).

RESULTS

LIPOSOME SIZE

All liposome formulations demonstrated log-normal distribution of particle size. The liposome sizes of the 7:2:1 (5% TA,+), 7:2:1 (10% TA,+), 7:2:1 (5% TA,-) and 7:2:1 (10% TA,-) were 17.5, 35.8, 2.8, and 2.0 μm, respectively. The particle sizes of the positively charged liposomes (17.5–35.8 μm) were approximately ten times larger than the negatively charged liposomes (2.0–2.8 μm). The smallest size (20 μm) was observed in the 7:2:1 (10% TA,-) liposome.

Table I
Equations for Calculations of Leakage or Degradation or Release Rate Constants, and Shelf Lives of TA in Liposomes[a]

Types of equation	Equations
Zero-order	$C_t = C_o + kt$
First-order	$\ln C_t = \ln C_o + kt$
Higuchi model	$C_t = C_o + kt^{1/2}$
Shelf life (first-order)	$t_{90} = 0.105 / k$

[a] C_o = initial concentration; C_t = concentration at given t; k = leakage or degradation or release rate constant; t = time; t_{90} = shelf life.

DETERMINATION OF TA CONTENT

The calibration graph of TA solution in DI water was demonstrated to be linear (r^2 = 0.9937), over the concentration range 4.0–20.0 μg/ml. The regression equation was as follows: y = 20.7903 x – 0.1942, where y is the concentration of TA (μg/ml) and x is the absorbance of the derivative of TA formed with 2,4,6-trinitrobenzosulfonic acid (mAU*s). The recoveries of TA from liposome formulations using spectrophotometric assay varied between 91.4% and 104.7%. The other components in liposome formulations neither reacted with the color reagent nor exhibited significant absorption at 415 nm.

STABILITY OF LIPOSOMES

The physical appearances of the 7:2:1 (5% TA,+), 7:2:1 (10% TA,+), 7:2:1 (5% TA,–), and 7:2:1 (10% TA,–) liposomes, either freshly prepared or stored at 4°, 30°, and 45° (± 1°)C for 90 days are shown in Table II. The physical notation of deterioration of liposome formulations, especially the positively charged liposome, can be observed from an increase in the turbidity of supernatant following storage at 30° and 45°C for 90 days (Table II). The amount of the remaining TA entrapped in various liposome formulations during storage at 4°, 30°, and 45°C for 90 days (Table III, Figure 1) was analyzed, using the proposed models (Table I), to estimate the leakage rate constant and shelf life of entrapped TA in liposomes (Table IV). The predicted shelf lives of TA in all liposome formulations varied between 19.0 and 65.1 days, with the longest shelf life observed in the 7:2:1 (10% TA,–) liposome (Table IV). The degradation rate constants of total TA in liposomes, and in the pellet of liposome formulations, estimated using the first-order equation, are presented in Table V. The degradation rate constants of total TA in all liposome formulations (0.0002–0.0016 d^{-1}) were about four to nine times lower than those in the pellet of liposomes (0.0018–0.0069 d^{-1}, Table V).

DRUG RELEASE

The cumulative percentages of the amount of TA released from various liposome formulations, 5% and 10% TA solutions in DI water during 24 hours at 37°C, are presented in Figure 1. The experimental data obtained were analyzed using the Higuchi model (Table I) to estimate the release rate constant of TA from liposomes (Table VI). The cumulative percentages of TA released from the positive and negative liposomes with the same drug concentration were not significantly different (Figure 2). However, high TA concentration in the liposomes appeared to exhibit a higher release rate constant of TA (Figure 1, Table VI). The total amount of TA released from liposome formulations during 24 h varied between 28.0% and 37.9%, with the highest amount observed in the 7:2:1 (10% TA,–) liposome (Table VI).

DISCUSSION

All liposome formulations appeared to be colloidal systems, either freshly prepared or stored at 4°, 30°, and 45°C for 90 days (Table II). No sedimentation and flocculation

Table II

Physical Appearances of Various Liposome Formulations With the Entrapped TA, Either Freshly Prepared or Stored at 4°, 30°, and 45°C for 90 Days

				Following storage for 90 days								
	Freshly prepared			Sedimentation			Flocculation			Turbidity		
Formulations	Sedimentation	Flocculation	Turbidity	4°C	30°C	45°C	4°C	30°C	45°C	4°C	30°C	45°C
7:2:1 (5% TA,+)	No	No	+1[a]	No	No	No	No	No	No	+1	+5	+4
7:2:1 (10% TA,+)	No	No	+1	No	No	No	No	No	No	+1	+5	+4
7:2:1 (5% TA,−)	No	No	+1	No	No	No	No	No	No	+1	+1	+3
7:2:1 (10% TA,−)	No	No	+1	No	No	No	No	No	No	+1	+1	+1

[a] +1 to +5 indicates an increase in the degree of turbidity of the supernatant.

Table III
Comparison of the Percentages of TA Remaining in Liposomes When Kept at 4± 1°, 30± 1°, and 45± 1°C for 0, 14, 30, 60, and 90 Days

	7:2:1 (5% TA,+)			7:2:1 (10% TA,+)			7:2:1 (5% TA,–)			7:2:1 (10% TA,–)		
Day(s)	4 ± 1°C	30 ± 1°C	45 ± 1°C	4 ± 1°C	30 ± 1°C	45 ± 1°C	4 ± 1°C	30 ± 1°C	45 ± 1°C	4 ± 1°C	30 ± 1°C	45 ± 1°C
0	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
14	86.48	88.95	85.47	99.3	92.85	100.77	102.2	98.62	93.8	105.04	102.92	102.92
30	84.01	80.81	82.85	93.17	94.19	90.24	77.82	80.17	73.42	88.3	94.81	93.93
60	90.99	79.65	63.37	88.83	68.47	70.52	70.66	68.04	70.11	83.85	85.16	76.83
90	68.6	56.4	52.62	77.09	64.77	62.22	81.54	58.54	67.77	83.48	86.99	77.92

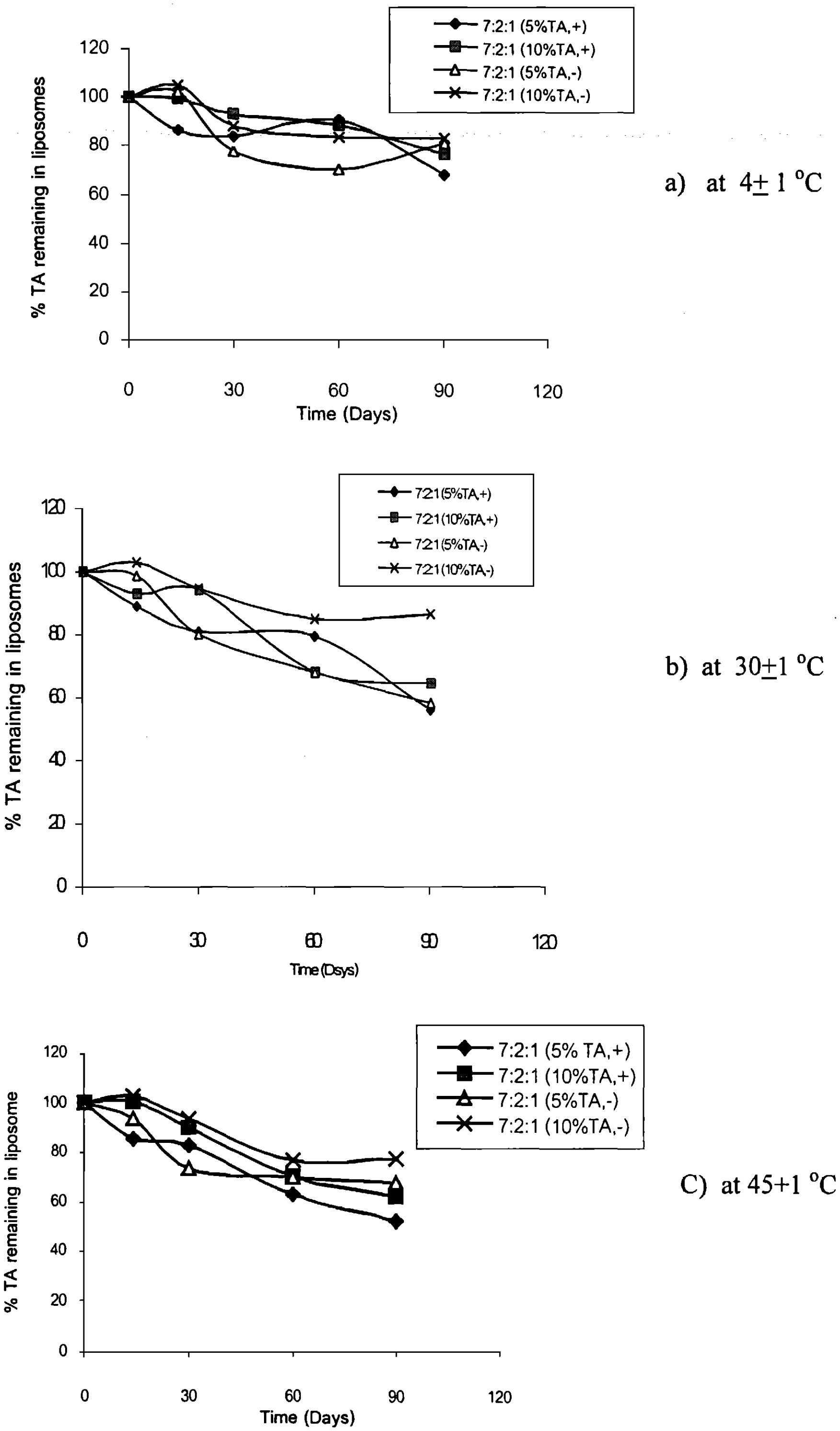

Figure 1. Comparison of the percentages of TA remaining in liposomes when kept at (a) 4 ± 1° (b), 30 ± 1°, (c) 45 ± 1°C for 0, 14, 30, 60, and 90 days. Symbols are the mean values of six determinations.

Table IV
Leakage Rate Constants (k) and Predicted Shelf Lives (t_{90}) of the Entrapped TA in Liposomes[a,b]

	k (day^{-1})			t_{90} (days)		
Formulations	4°C	30°C	45°C	4°C	30°C	45°C
7:2:1 (5% TA,+)	0.0030	0.0046	0.0055	35.1	22.8	19.0
7:2:1 (10% TA,+)	0.0030	0.0039	0.0046	35.1	26.8	22.8
7:2:1 (5% TA,−)	0.0023	0.0032	0.0039	45.6	32.6	26.8
7:2:1 (10% TA,−)	0.0016	0.0023	0.0030	65.1	45.6	35.1

[a] Experimental data represent the mean value of six determinations.
[b] The k and t_{90} values were estimated using the first-order equation.

Table V
Degradation Rate Constants (k in day^{-1}) of Total TA in Liposomes, and in the Pellets of Liposomes[a,b]

	k of total TA in liposome			k of total TA in pellet		
Formulations	4°C	30°C	45°C	4°C	30°C	45°C
7:2:1 (5% TA,+)	0.0002	0.0009	0.0016	0.0030	0.0055	0.0069
7:2:1 (10% TA,+)	0.0005	0.0016	0.0014	0.0028	0.0053	0.0058
7:2:1 (5% TA,−)	0.0002	0.0007	0.0009	0.0062	0.0062	0.0023
7:2:1 (10% TA,−)	0.0014	0.0012	0.0012	0.0023	0.0018	0.0035

[a] Experimental data represent the mean value of six determinations.
[b] The k and t_{90} values were estimated using the first-order equation.

Table VI
Release Rate Constants (k) of TA and the Total Amount of TA Released From Liposomes and Solution During 24 Hours at 37°C[a,b]

Formulation	Release rate constant ($\%/h^{1/2}$)	Total release of TA (%)
7:2:1 (5% TA,+)	6.12 ± 0.06	28.05 + 1.45
7:2:1 (10% TA,+)	8.13 ± 0.05	36.20 ± 1.24
7:2:1 (5% TA,−)	6.29 ± 0.07	29.22 + 1.71
7:2:1 (10% TA,−)	8.27 ± 0.04	37.93 ± 1.04
5% TA solution	18.89 ± 0.21	81.69 ± 5.03
10% TA solution	19.77 ± 0.20	87.32 ± 4.90

[a] Experimental data represent the mean value of six determinations.
[b] The values were estimated using the Higuchi model equation.

were observed in all liposome formulations following storage at 4°, 30°, and 45°C for 90 days, indicating the high physical stability of these liposome systems (Table II). These formulations may have stable electrically double layers of the multilamellar vesicles, within the pH range of 6.9 to 7.9. Following storage at 30° and 45°C for 90 days, the positively charged liposomes with the entrapped TA demonstrated a more marked increase in the turbidity of the supermatant, compared to the negatively charged liposomes with the entrapped TA (Table II). Thus, charges appeared to affect the physical stability of the liposomes.

In all stability studies, better correlations (r^2 = 0.91) were obtained when the observed

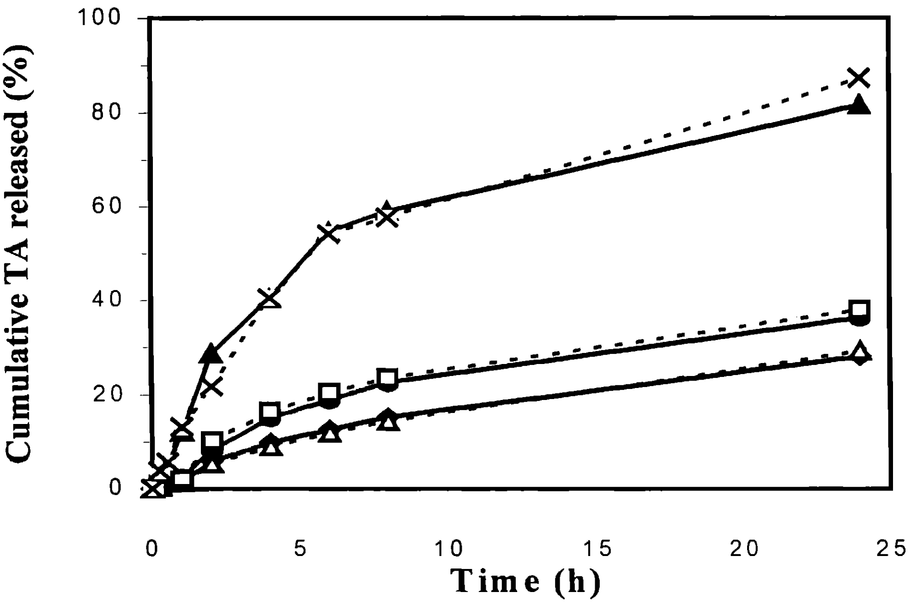

Figure 2. Cumulative percentages of TA released from various liposome formulations and solutions during 24 h at 37°C: ♦, 7:2:1 (5% TA,+); ●, 7:2:1 (10% TA,+); △, 7:2:1 (5% TA,–); 7:2:1 (10% TA,–); ▲, 5% TA solution; X, 10% TA solution. Symbols are the mean values of six determinations.

data were fitted to the first-order equation, compared to those fitted to the zero order and Higuchi model equations. Thus, the first-order equation seemed to be the most acceptable approach to be employed in this stability study, and was used to estimate the leakage rate constant and shelf life of the entrapped TA in various liposome formulations. On the other hand, for the release study, the Higuchi model exhibited the best fit (r^2 = 0.97) when the experimental data obtained were fitted to the proposed model equations (Table I). Thus, this model was used to estimate the release rate constant of TA from various liposome formulations and TA solutions (5% and 10%) in DI water.

The leakage rates of entrapped TA in the positively charged liposomes (0.0030–0.0055 d^{-1}) were higher than those in the negatively charged liposomes (0.0016–0.0039 d^{-1}, Table VI). The negatively charged liposome with the entrapped 10% drug {7:2:1(10% TA,–)} demonstrated the slowest leakage rate (Table VI). This may be associated with the positive charges of TA, which may bind strongly with the negative charges of the liposomal membrane, thereby lowering the leakage rate of the entrapped drug from the liposomes (21). Thus, charges appeared to affect the leakage rate and shelf life of TA in liposomes. In contrast, the degradation rate constants of total TA in the positive and negative liposomes, and in the pellets of positive and negative liposomes, were broadly comparable (Table V).

The leakage rate was accelerated as the storage temperature increased (Table VI). The best storage condition was 4°C, since the lipid becomes more flexible and fluid at high temperature (22). Thus, the entrapped drug can leak easily from the lipid bilayers of liposomes. In consideration of the degradation rate constants of total TA in liposomes (Table V), TA in all liposome formulations exhibited relatively high stability, as more than 90% of the drug remained following storage at 4°, 30°, and 45°C for 60 days. This indicated the stability of TA when entrapped in liposomes. Similar to the leakage rate constants, the degradation rate constants of total TA in liposomes and in the pellets of liposomes were accelerated as the storage temperature increased (Table V).

In the *in vitro* release study, the release rate constants of TA from the 7:2:1(5% TA,+) and 7:2:1 (5% TA,–) liposomes (6.1–6.3 %/$h^{1/2}$) were lower than the 7:2:1(10% TA,+) and 7:2:1 (10% TA,–) liposomes (8.1–8.6 %/$h^{1/2}$, Table VI). Thus, the release rates of TA from liposome formulations were dependent on the initial TA concentrations and the hydrophilicity of TA. TA is a hydrophilic drug with weak positive charge properties that can be incorporated in an aqueous layer of the multilamellar liposomes. In the negatively charged liposomes {7:2:1 (5% TA,–) and 7:2:1 (10% TA,–)}, TA may interact with the surface charge of liposomal bilayers, resulting in the difficulty of TA to be released from liposomes (23,24). However, there were no significant differences in the release rates of TA from the positively and negatively charged liposomes (6.1–8.1 %/$h^{1/2}$ vs 6.3–8.6 %/$h^{1/2}$, Table VI). Thus, the release rates of TA from liposomes were apparently independent of charges.

The release rates of TA from the 7:2:1 (5% TA,+), 7:2:1 (10% TA,+), 7:2:1 (5% TA,–), and 7:2:1 (10% TA,–) varied between 6.1 and 8.6%/$h^{1/2}$. In comparison, the respective release rates of TA from 5% and 10% solutions in DI water were 18.9% and 19.8%/$h^{1/2}$. Thus, the release rates of TA from all liposome formulations were ~ 3 times slower than that from the solution. This suggests that all liposome formulations prolong and sustain the release of TA.

CONCLUSION

TA in all liposome formulations demonstrated relatively high chemical stability. The best formulation, evaluated from liposomal size, physical stability, leakage rate constant, shelf life, release rate, and the total amount of TA released, was found to be the negatively charged liposome with the 10% entrapped TA {7:2:1 (10% TA,–)}. In addition, TA entrapped in liposome formulations showed sustained and prolonged release, as their release rates were slower than that in the solution. We have previously found that the 7:2:1 (10% TA,–) liposome exhibited a relatively high percentage of entrapment of the drug (13%). The charges of liposomes appeared to affect the physical stability, leakage rate, and shelf life of TA in liposomes, whereas TA concentrations apparently affect the release rate of TA and the total amount of TA released from liposomes. It is a challenge for the future development of TA entrapped in liposomes, for parenteral and topical applications. The developed topical product of TA can function not only as a depot system but as a product with low irritation as well.

ACKNOWLEDGMENTS

The authors acknowledge Dr. Kuncoro Foe for his assistance in the preparation of the manuscript. We also acknowledge JJ-Degussa (T) Ltd., Thailand, for the gift of Emulmetik 950®, and Thistle Corp., Ltd. (Thailand) for the gift of tranexamic acid. We thank the Research and Development of Natural Products for Thai Traditional Medicines Research Unit, Pharmaceutical-Cosmetic Raw Materials and Natural Products Research and Development Center, Institute for Science and Technology Research and Development at Chiang Mai University, for financial support.

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